CLINICAL · LARYNX & AIRWAY

Lasers, radiofrequency, cryosurgery and hyperbaric oxygen

Lasers, radiofrequency, coblation, cryotherapy and hyperbaric oxygen are tools, and each works through one physical principle. Know which tissue absorbs which wavelength, how much collateral damage each tool leaves, and the safety rules. The airway fire is the disaster every ENT surgeon working with lasers must prevent.

AT A GLANCE

- CO2 (10,600 nm) is absorbed by water; KTP by haemoglobin; Nd:YAG coagulates deeply.
- CO2 cannot pass through a fibre.
- Airway fire: stop laser, stop gas, remove tube, saline, reintubate, bronchoscopy.
- Coblation works at 40 to 70 degrees through a saline plasma.

1 CHAPTER ONE Lasers

Light amplification by stimulated emission of radiation.

1.1 How a laser works

1. **Pumping:** an energy source (electric current or a flash lamp) excites the atoms of an *active medium*. The medium can be a gas (CO2, argon, helium–neon), a solid crystal (Nd:YAG, KTP, erbium, holmium), a liquid dye, or a semiconductor (diode).
2. **Spontaneous and stimulated emission:** excited electrons fall back and release photons. A photon striking another excited atom makes it release an identical photon, of the same wavelength, direction and phase. This is *stimulated emission*.
3. **Population inversion:** when more atoms are excited than not, amplification begins.
4. **Resonance:** two mirrors at the ends of a resonant cavity (one fully reflecting, one partly) bounce the light back and forth to amplify it. The beam leaves through the partly reflecting mirror.

The result is light that is *monochromatic*, *coherent* (in phase), *collimated* (parallel) and of high intensity, and that can be focused to a tiny spot.

1.2 Laser and tissue

A laser beam meeting tissue is *reflected*, *transmitted*, *scattered* or *absorbed*, and only the absorbed energy has any effect. Absorption depends on the **chromophore**, the target molecule that takes up a particular wavelength. *Water* absorbs infrared, which is why the CO2 and Er:YAG lasers cut so cleanly. *Haemoglobin* absorbs green and yellow light, with peaks at 418, 542 and 577 nm, the target of the KTP laser at 532 nm and the pulsed dye laser at 585 to 595 nm. *Melanin* absorbs across the spectrum, mainly in the visible and near-infrared, the target of ruby, alexandrite and Nd:YAG lasers.

The absorbed energy acts as heat. At about 60°C proteins denature and coagulate reversibly, around 80°C collagen breaks down, and at 100°C water boils and the tissue *vaporises*. A laser wound therefore has three zones:

1. A crater of vaporisation.
2. A zone of thermal necrosis, which seals small vessels, nerves and lymphatics.
3. A zone of thermal conductivity and repair, which is reversible.

Depth of penetration follows from absorption. Longer visible wavelengths penetrate further, but water-absorbed infrared light from CO2 and Er:YAG lasers penetrates only microns, giving precise cuts with little collateral damage. The Nd:YAG laser, by contrast, coagulates about 4 mm deep, which is good for haemostasis but imprecise. Beyond heat, lasers act photochemically in photodynamic therapy, photoacoustically in stone fragmentation, and by photoablation in the excimer laser.

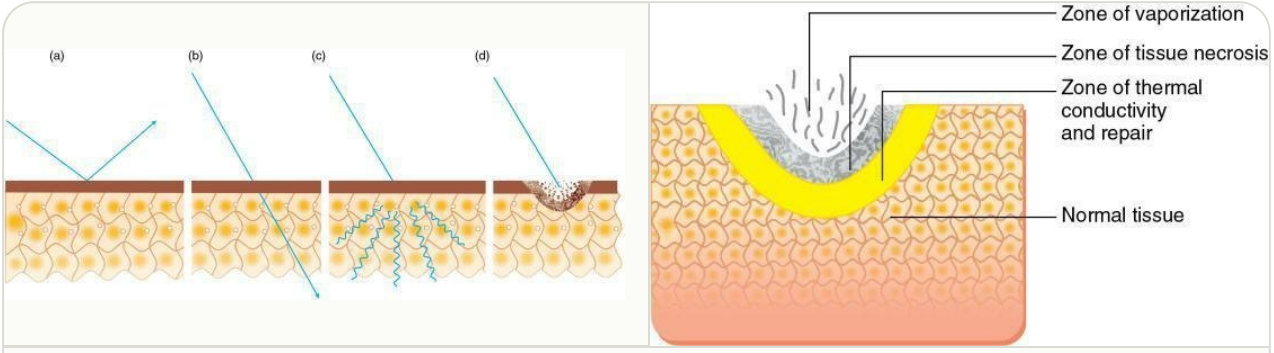


FIGURE 1 First, the fates of a laser beam in tissue: (a) reflection, (b) transmission, © scatter, (d) absorption with tissue destruction. Second, the zones of tissue damage: vaporisation, thermal necrosis, and thermal conductivity and repair.

Dhingra, figs 71.1 and 71.2

1.3 Parameters the surgeon controls

The surgeon sets the **power** in watts and the **spot size**, which is smallest at the focal point: a focused beam cuts, while a defocused beam coagulates and ablates. Exposure can be **continuous**, **pulsed**, or **superpulsed and Q-switched**, in which very short pulses are separated by intervals longer than the tissue's *thermal relaxation time*, so that heat dissipates between pulses and collateral damage falls. **Power density** (W/cm²), the power divided by the spot area, sets the rate of cutting, and **fluence** (J/cm²), the power density multiplied by time, is the total energy delivered.

1.4 Delivery

Visible and near-infrared lasers, the argon, KTP, Nd:YAG, diode and holmium lasers, travel down *flexible optical fibres* and can be used through endoscopes and bronchoscopes. The CO2 laser cannot pass through ordinary fibres; it travels by an *articulated arm* with mirrors to a micromanipulator on the operating microscope, or through hollow flexible waveguides. Because CO2 and Nd:YAG beams are invisible, a red helium–neon *aiming beam* shows where they will strike.

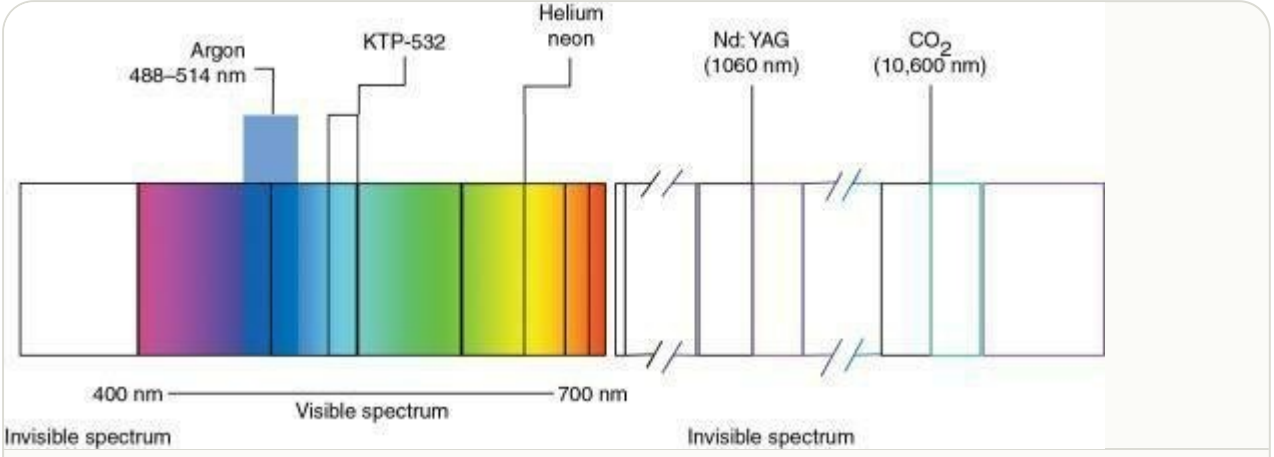


FIGURE 2 The electromagnetic spectrum, with the wavelengths of the common surgical lasers. Dhingra, fig. 71.3

COMMON LASERS IN ENT

LASER	WAVELENGTH	ABSORBED BY	DELIVERY	TYPICAL USES
CO2	10,600 nm (far infrared, invisible)	Water	Articulated arm, micromanipulator, waveguide	Workhorse of laryngology: nodules, polyps, cysts, papillomas, Reinke oedema, leukoplakia, T1 glottic cancer (transoral laser microsurgery), cordotomy and arytenoidectomy, stapedotomy, skin resurfacing
KTP	532 nm (green)	Haemoglobin, melanin	Fibre	Vascular lesions, papillomas (office angiolytic laser), stapedotomy, sinus surgery, HHT
Argon	488 to 514 nm (blue–green)	Haemoglobin, melanin	Fibre	Vascular skin lesions, stapes surgery (historical)
Nd:YAG	1064 nm (near infrared, invisible)	Pigment; deep penetration	Fibre	Deep coagulation: palliation of tracheobronchial and oesophageal tumours, vascular malformations, HHT
Diode	800 to 980 nm	Haemoglobin, melanin	Fibre	Turbinate reduction, stapedotomy, dacryocystorhinostomy, vascular lesions
Er:YAG	2940 nm	Water (peak)	Fibre or arm	Fine skin resurfacing, bone (stapes)
Ho:YAG	2100 nm	Water	Fibre	Bone, sinus, stones
Pulsed dye	585 to 595 nm (yellow)	Haemoglobin	Fibre	Port-wine stains, haemangiomas, telangiectasia

1.5 Advantages and disadvantages

Lasers offer precision, haemostasis and a bloodless field, cause little postoperative oedema and pain, and reach targets through endoscopes. Against this are their cost, the training they demand, thermal injury, the lack of tactile feedback, and the hazards of *fire and eye injury*.

1.6 Laser safety

Safety starts with organisation: a laser safety officer, trained and certified staff, and a "*laser in use*" sign on a closed door.

Everyone in the room, including the patient, wears *wavelength-specific goggles*: clear glass or plastic for CO2, orange or amber for argon and KTP, and green-blue for Nd:YAG. The reason is anatomical. Visible and near-infrared lasers (400 to 1400 nm) pass through the transparent media and burn the *retina*, whereas CO2 and other far-infrared lasers are absorbed by water at the front of the eye and burn the *cornea*. The patient's eyes are covered with wet pads and exposed skin with wet towels. The smoke plume can carry viable viral particles, such as HPV from papillomas, and is mutagenic, so a separate smoke evacuator is used.

The **airway fire** needs the three elements of the *fire triangle*: heat from the laser, fuel from the tube, drapes and cottonoids, and an oxidiser, oxygen or nitrous oxide. It is prevented by removing each element. Use laser-safe tubes, metal or specially wrapped, or jet ventilation without a tube; fill the cuff with *saline coloured with methylene blue* so that a puncture shows at once; protect the cuff and tissues with wet cottonoids; keep oxygen at 30% or less, and never above 40%; use *no nitrous oxide*; use the lowest effective power; and direct the beam only at the target.

AIRWAY FIRE DRILL

1. Stop the laser.
2. Stop ventilation and disconnect oxygen.
3. Remove the tube.
4. Pour saline into the airway.
5. Ventilate by mask with air.
6. Reintubate.
7. Do bronchoscopy to assess the injury and remove debris.

8. Give steroids, antibiotics and close follow-up for stenosis.

OPERATIVE GUIDES IN THEATRE 2 operations

Transoral CO2 laser microsurgery of the larynx and the tongue base.

Transoral CO2 Laser Microsurgery for Cancer of the Larynx
Transoral laser microsurgery
Transoral Laser Microsurgery of the Base of the Tongue
TLM tongue base

1.7 Photodynamic therapy

In photodynamic therapy a photosensitiser, given systemically (porphyrins, temoporfin) or topically to the skin (aminolaevulinic acid), concentrates in tumour cells. Light of a matching wavelength, around 630 to 652 nm, then generates oxygen free radicals that kill them. It is used for superficial cancers of the skin, oral cavity and larynx, for field cancerisation, and to palliate recurrences, and patients must avoid sunlight for weeks because the skin is photosensitive.

2 CHAPTER TWO Radiofrequency and coblation

Lower temperatures.

2.1 Radiofrequency ablation

Radiofrequency ablation passes a high-frequency alternating current, about 350 kHz to 4 MHz and commonly 460 kHz, through a probe placed *within* the tissue. Ionic agitation heats the tissue to about 70 to 85°C, coagulating it without charring, and fibrosis over the following three to six weeks shrinks it.

It reduces the volume of the inferior turbinates, the soft palate and the tongue base in snoring and obstructive sleep apnoea, and is used for tonsil reduction, lingual thyroid, thyroid nodules and microcystic lymphatic malformations. At the tongue base the hypoglossal neurovascular bundle lies about 2.7 cm deep and 1.6 cm lateral

to the foramen caecum, and must be respected. The procedure is done as an outpatient, causes little pain and can be repeated.

2.2 Coblation (controlled ablation)

Coblation passes radiofrequency energy through *conductive saline* to create a thin *plasma field* of energised sodium ions. These break molecular bonds and dissolve tissue at only *40 to 70°C*, against 400 to 600°C for electrocautery, so there is little charring or collateral injury. A single wand ablates, suctions, irrigates and coagulates.

Coblation is used for tonsillectomy and adenoidectomy, including intracapsular tonsillotomy in children; tongue base and palatal surgery; turbinates and nasal polyps; laryngeal papillomas and cysts; cordotomy; subglottic and tracheal lesions; and juvenile angiofibroma. After coblation tonsillectomy, pain and haemorrhage are similar to or slightly better than other hot techniques, and intracapsular coblation causes less pain and bleeding.

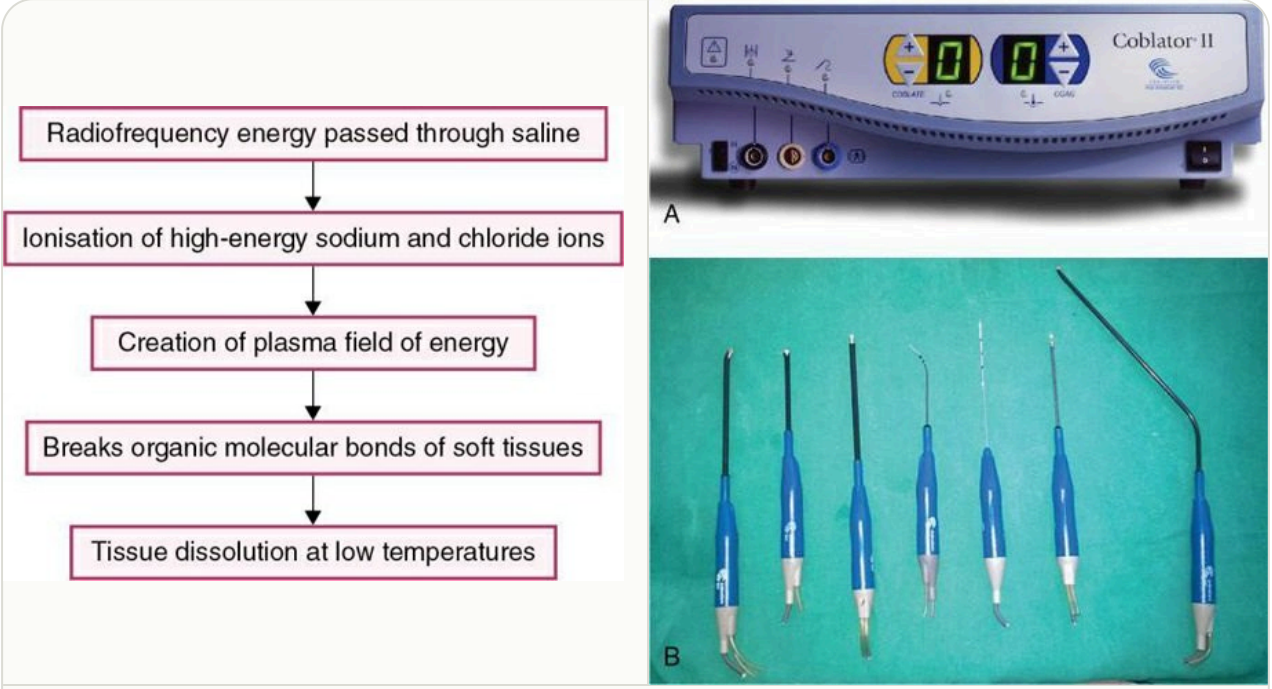


FIGURE 3 First, the principle of coblation. Second, (A) a coblation generator and (B) the different wands.

Dhingra, figs 71.5 and 71.6

COBLATION COMPARED WITH ELECTROSURGERY

FEATURE	COBLATION	ELECTROSURGERY (DIATHERMY)
Temperature	40 to 70°C	400 to 600°C
Mechanism	Chemical: plasma breaks molecular bonds	Thermal: heating, charring
Collateral damage	Superficial, minimal	Deeper
Field	Saline irrigated, suction built in	Dry

3 CHAPTER THREE Cryosurgery

Destruction by freezing.

3.1 Principle

Cryosurgery uses liquid nitrogen, by spray or probe at down to −196°C, or nitrous oxide and CO2 probes, which rely on the *Joule–Thomson effect*: compressed gas expanding through a small orifice cools the tip to about −70°C or below. The tissue is frozen rapidly to −20 to −30°C or colder, *thawed slowly*, and the cycle repeated, with a margin of normal tissue and, if needed, a thermocouple to monitor temperature.

Cells die by several mechanisms: ice crystals and *dehydration* leave toxic concentrations of electrolytes; membrane lipoproteins are *denatured*; thermal shock damages the cells; and *vascular stasis* with microthrombosis causes ischaemic necrosis, which is why vascular lesions respond so well. There may also be an immune "cryoimmunisation" effect. The slough separates in three to six weeks and heals by secondary intention with little scarring.



FIGURE 4 Cryotherapy for a haemangioma of the cheek.
Dhingra, fig. 72.1

3.2 Uses, advantages and limits

Cryotherapy treats haemangiomas and vascular malformations, premalignant lesions such as leukoplakia and actinic keratosis, and superficial skin cancers such as Bowen's disease and small basal cell carcinomas, with cure rates above 90%, including lesions over cartilage, which resists freezing. It also palliates advanced and recurrent tumours, reduces turbinates, and was formerly used for the tonsils of high-risk patients.

It can be done as an outpatient, often without anaesthesia, is safe in bleeding disorders and poor-risk patients, causes little pain, leaves minimal scarring and no keloid, and is cheap and repeatable. Its limits are that it provides no histology or margin control, its depth is imprecise, it leaves hypopigmentation and hair loss in skin, it causes swelling and slough, and it is unsuitable for recurrent or ill-defined cancers.

4 CHAPTER FOUR Hyperbaric oxygen

Pressure and oxygen.

4.1 Principle

Breathing 100% oxygen at more than one atmosphere absolute, usually 2.0 to 2.5 ATA for 90 to 120 minutes over a series of 10 to 40 sessions, increases the oxygen dissolved in plasma several-fold. This raises tissue oxygen tension, stimulates angiogenesis and fibroblast activity, improves leucocyte killing, reduces oedema and shrinks gas bubbles.



FIGURE 5 A multiplace hyperbaric chamber.
Dhingra, fig. 71.4

4.2 ENT uses

For **osteoradionecrosis and soft-tissue radiation injury**, hyperbaric oxygen is an adjunct to debridement and reconstruction, but the evidence is mixed: Marx's older protocol supported giving it before dental extraction in irradiated jaws, whereas the randomised HOPON trial found no reduction in osteoradionecrosis, and pentoxifylline–tocopherol is an alternative. It is also used for refractory **necrotising**

otitis externa and skull base osteomyelitis, for compromised flaps and grafts, and for **sudden sensorineural hearing loss** as salvage or with steroids within two weeks of onset in moderate to profound loss. Outside ENT it treats decompression sickness, including inner ear decompression sickness in divers, air embolism, carbon monoxide poisoning and necrotising fasciitis.

4.3 Complications and contraindications

Middle ear barotrauma is the commonest complication and may need grommets; sinus barotrauma, oxygen toxicity seizures, temporary myopia and claustrophobia also occur. An untreated pneumothorax is an absolute contraindication. Eustachian tube dysfunction, upper respiratory infection, emphysema with bullae, uncontrolled epilepsy and certain chemotherapy drugs such as bleomycin and doxorubicin are relative contraindications.

5 CHAPTER FIVE Narrow band imaging

Seeing the vessels.

5.1 Principle and value

Narrow band imaging filters the endoscope's light into two bands, *415 nm (blue)* and *540 nm (green)*, which are absorbed by haemoglobin and penetrate only the superficial layers, so superficial capillaries appear brown and submucosal veins cyan. Abnormal *intraepithelial papillary capillary loops*, dilated and tortuous with brown dots, mark dysplasia and early cancer. This helps to detect small cancers under 1 cm and second primaries, to target biopsies, to define resection margins and to survey for recurrence after radiotherapy. Its accuracy is reduced by secretions, bleeding, keratosis and post-radiation change.

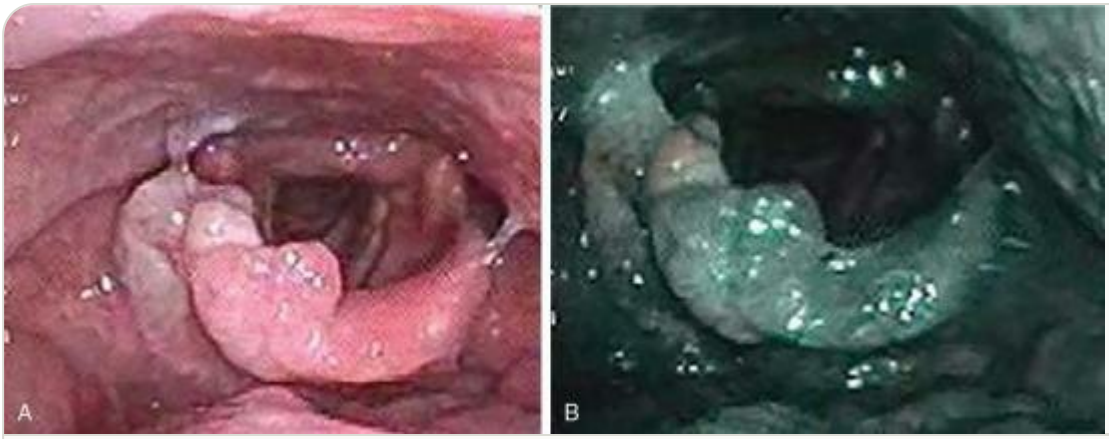


FIGURE 6 Carcinoma of the epiglottis (A) under white light and (B) under narrow band imaging.
Dhingra, fig. 71.7

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **What LASER stands for,** the properties of laser light, and its components.
- **Chromophores;** CO2 versus KTP versus Nd:YAG: wavelength, absorption, delivery, uses.
- **The three zones of a laser wound;** power density and fluence.
- **Laser safety:** eye protection by wavelength, laser-safe tubes and the blue-dyed saline cuff, low oxygen and no nitrous oxide, the airway fire drill.
- **Coblation versus diathermy;** radiofrequency for sleep surgery.
- **Mechanisms of cryotherapy;** freeze fast, thaw slowly.
- **Hyperbaric oxygen:** how it works, ENT indications, complications.

6.2 Common pitfalls

COMMON PITFALLS

- Using nitrous oxide or high oxygen during laser airway surgery.

- Giving the theatre team the wrong goggles for the wavelength.
- Treating an undiagnosed lesion with cryotherapy or laser vaporisation without taking a biopsy first.
- Forgetting ear barotrauma in patients starting hyperbaric oxygen.

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IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.